

APS110 Tutorial #2

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Practice Question 1

1. A cylindrical rod with a diameter of 12.5 mm and a length of 600 mm is loaded in tension. If the rod is expected to not plastically deform or elongate more than 1.4mm under an applied load of 29 kN, which of the four metals can be used for this application? A list of materials can be found below.

Material	Young's Modulus (GPa)	Yield Strength (MPa)	Tensile Strength (MPa)
Aluminum	70	255	420
Brass	100	345	420
Copper	110	210	275
Steel	207	450	550

Practice Question 1

$$\sigma = F/A = 29000 / \pi(12.5e-3 / 2)^2 \text{ [N/m}^2\text{]} = 236 \text{ MPa} > 210 \text{ MPa}$$

=> cannot use copper

$$\varepsilon = \Delta l / l_0, \sigma = E\varepsilon \Rightarrow \Delta l = \sigma l_0 / E$$

$$\Delta l_{\text{Al}} = (236e6 \cdot 600e-3) / 70e9 \approx 2.02 \text{ mm} > 1.4 \text{ mm}$$

$$\Delta l_{\text{Brass}} = (236e6 \cdot 600e-3) / 100e9 \approx 1.42 \text{ mm} > 1.4 \text{ mm}$$

$$\Delta l_{\text{Steel}} = (236e6 \cdot 600e-3) / 207e9 \approx 0.68 \text{ mm} < 1.4 \text{ mm}$$

∴ Only steel can be used

Practice Question 2

2. Sketch a figure for the net interatomic forces versus interatomic spacing between two atoms. Clearly label the axes and indicate the shape of the curve for the attractive forces, repulsive forces, and net force.
 - a) Indicate the source of each of these forces and discuss the significance of the point where the net force curve intersects the horizontal axis
 - b) Explain the relationship of Young's Modulus to this curve and why it is independent of geometry

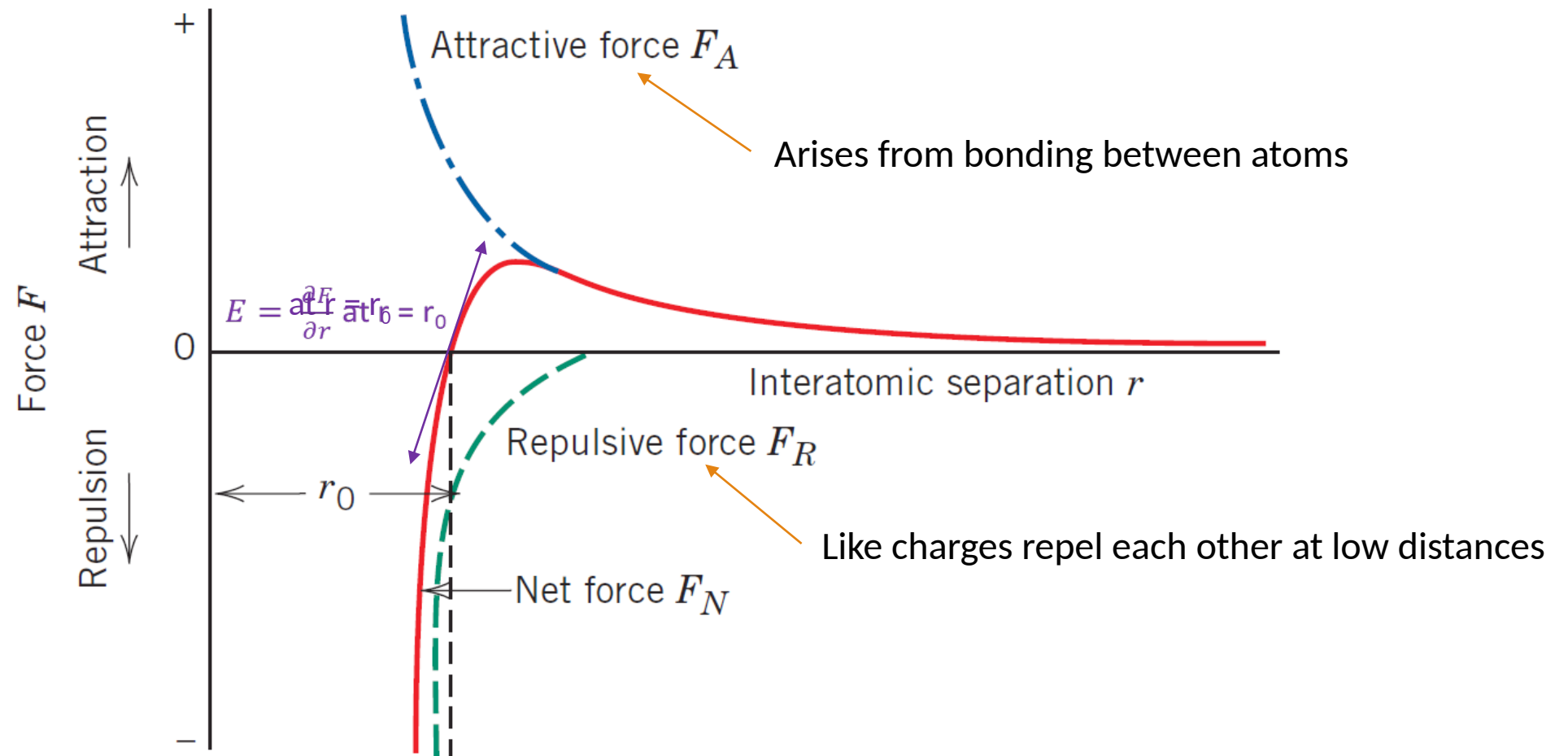


Figure from "Fundamentals of Materials Science and Engineering", 4th ed., Callister

Practice Quiz

1. During a tensile test of a given material, Young's modulus depends on:
 - a) The length of the sample
 - b) The diameter of the gauge region
 - c) The angle of the sample's shoulders
 - d) None of the above ☐ Young's modulus is independent of sample geometry (see previous slide)
2. A component you are designing must withstand an applied stress of 900 MPa. The dimensions of the component are: 12mm width, 35 mm height, and 67 cm length. Assuming you apply the load along the long axis, what is the maximum load associated with this applied stress?
3. A tensile stress is to be applied along the long axis of brass tensile specimen that has an initial gauge length $l_0 = 30$ mm. The whole tensile specimen is 2.0mm thick, with the grip region being 1.2cm wide and the reduced section half that width. Determine the magnitude of the load required to produce a 0.15 mm change in the gauge length if the deformation is entirely elastic. (brass modulus $E = 97$ GPa)
4. Consider a person attempting to pop a balloon with a series of nails, assuming they are only able to apply a maximum force of 100N with their arms. A balloon typically bursts when a pressure of 110kPa is applied to its surface. Considering that they could effortlessly pop the balloon with a single nail when applying 5.5N, what is the minimum number of nails on the wooden board for this experimenter to be completely incapable of bursting this balloon? (Consider that the pressure is distributed equally among the nails)
5. A hypothetical metal has a Young's modulus of 200 GPa and a yield strength of 600 MPa. This cylindrical sample with a diameter of 10mm and a length of 500mm has elongated to a final length of 506 mm. Is this bar undergoing elastic or plastic deformation? If this were purely elastic deformation, what is the maximum length that the sample could elongate to?

2. $\sigma = F/A \Rightarrow F = \sigma A = (900 \times 10^6) \cdot (12 \times 10^{-3} \cdot 35 \times 10^{-3}) [\text{Pa} \cdot \text{m}^2] = 378 \text{ kN}$

3. $\varepsilon = 0.15 \text{ mm} / 30 \text{ mm} = 0.005$

$$\sigma = E\varepsilon = 97 \times 10^9 \cdot 0.005 = 485 \text{ MPa}$$

$$F = \sigma A = 485 \times 10^6 \cdot [2 \times 10^{-3} \cdot (1.2 \times 10^{-3} / 2)] = 5.82 \text{ kN}$$

4. $A_{\text{nail tip}} = 5.5 / 110 \times 10^3 [\text{N/Pa}] = 5 \times 10^{-5} \text{ m}^2$

$$\sigma \propto 1/n, \text{ where } n \equiv \text{number of nails, so } \sigma = F/nA \Rightarrow n = F/\sigma A = 100 / (110 \times 10^3 \cdot 5 \times 10^{-5}) = 18.1 \approx 19 \text{ nails}$$

5. Sample is only elastic up to σ_y , so $\varepsilon_{\text{elastic}} = \sigma_y/E = 600 \times 10^6 / 200 \times 10^9 = 0.003$

$$\varepsilon = (l_i - l_0) / l_0 \Rightarrow l_i = l_0 \varepsilon + l_0 = (0.003 \cdot 500) + 500 = 501.5 \text{ mm} < 506 \text{ mm}$$

$\therefore$ The sample is undergoing plastic deformation, as the maximum elastic elongation is 1.5 mm